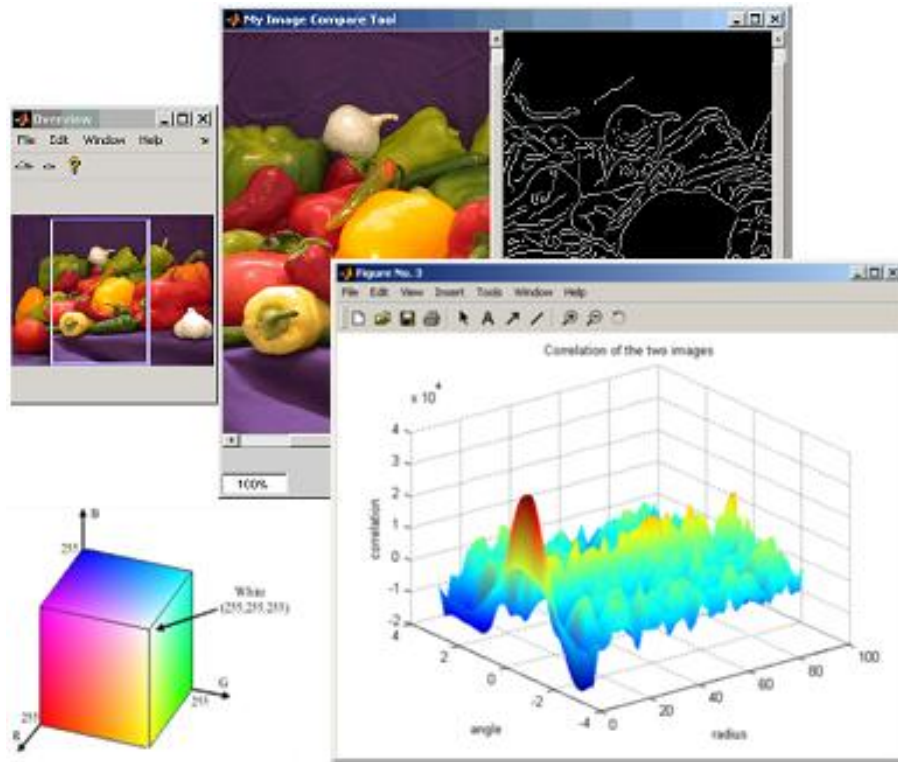


Digital Image Processing

LECTURE 9: HISTOGRAM PROCESSING



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Image Enhancement (Histogram Processing)

Histogram

Histogram is a discrete function formed by **counting the number of pixels that have a certain gray level** in the image.

OR

Histogram is the graphical representation of **Frequency of occurrence of each gray level** in the image

Histogram Representation Methods

❑ Method 1:

Un-normalised Histogram

- Histogram of a digital image with gray levels in the range $[0, L-1]$ is a discrete function

$$h(r_k) = n_k$$

Where,

- r_k : the k th gray level
- n_k : the number of pixels in the image having gray level r_k
- $h(r_k)$: histogram of a digital image with gray levels r_k

Histogram Representation Methods

□ Method 2:

Normalised Histogram

- Dividing each of histogram at gray level r_k by the total number of pixels in the image, N

$$p(r_k) = n_k / N$$

Where,

- r_k : be the input gray intensity level
- N : be the total number of pixels in an image
- $k = 0, 1, \dots, L-1$
- $p(r_k)$ gives an estimate of the probability of occurrence of gray level r_k

- The sum of all components of a normalized histogram is equal to 1

Numerical Examples

Ques1: Plot the (a) Un-normalised and (b) Normalised Histogram for the following digital image.

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = [0,9]

Histogram Representation Methods

Method 1: Un-normalised Histogram

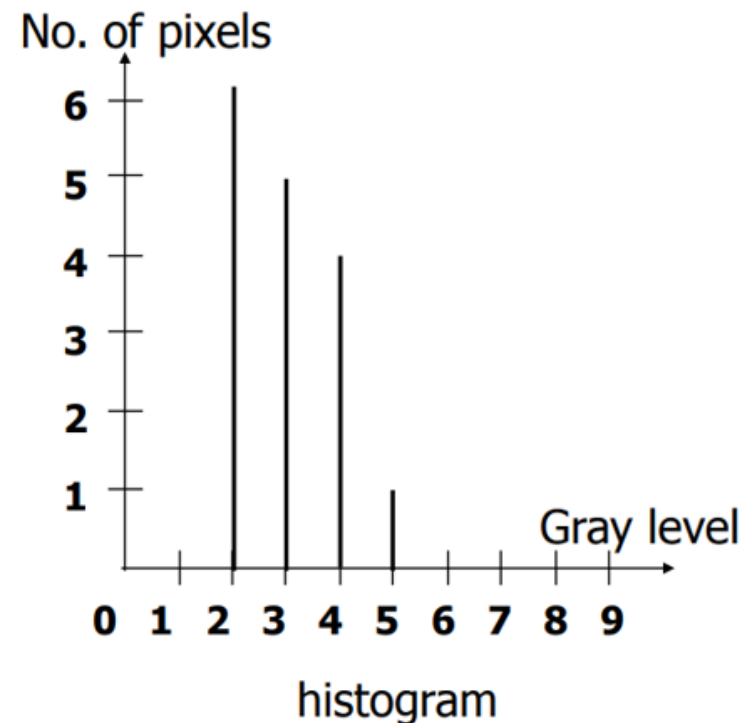
- **x-axis:** gray levels
- **y-axis:** no of pixels in the image in each gray levels.

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = [0,9]

r_k	n_k
0	0
1	0
2	6
3	5
4	4
5	1
6	0
7	0
8	0
9	0



Histogram Representation Methods

❑ Method 2: **Normalised Histogram**

- **x-axis:** gray levels
- **y-axis:** probability of occurrence of no of pixels in each gray levels

Intensity value (r_k)	0	1	2	3	4	5	6	7	8	9
Number of Pixels (n_k)	0	0	6	5	4	1	0	0	0	0
n_k/N	0	0	0.38	0.31	0.25	0.06	0	0	0	0

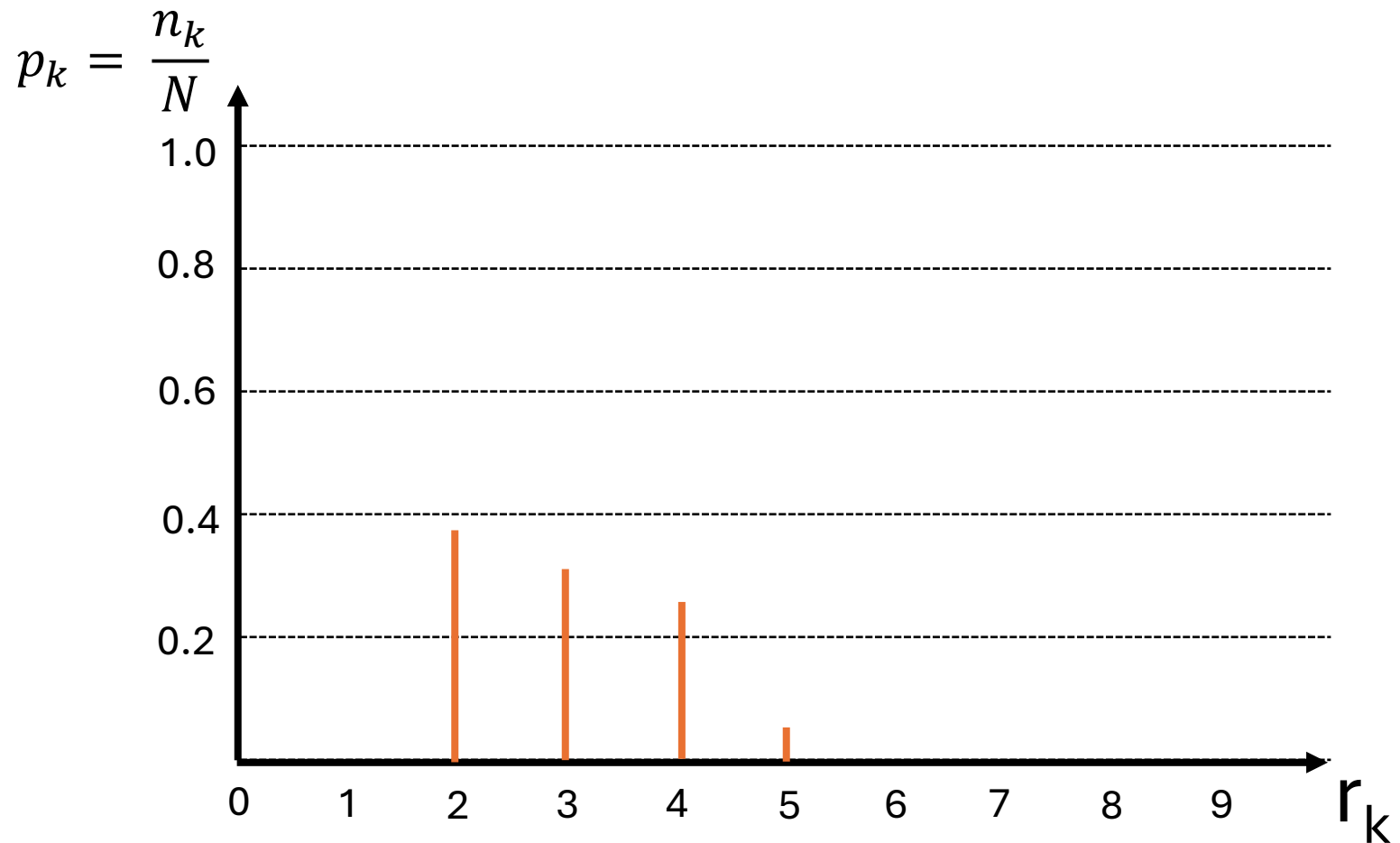
2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = $[0,9]$

Histogram Representation Methods

Intensity value (r_k)	0	1	2	3	4	5	6	7	8	9
Number of Pixels (n_k)	0	0	6	5	4	1	0	0	0	0
n_k/N	0	0	0.38	0.31	0.25	0.06	0	0	0	0



Numerical Examples

Ques 2: Construct the histogram for the image shown below:

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0	0	0	4	6
3	3	3	2	1
5	5	3	2	2
2	2	0	1	1
4	4	3	1	3

Numerical Examples

Ques 3: An image with gray levels between 0 to 7 is given below. Find the histogram of an image and normalize it

1	6	2	2
1	3	3	3
4	6	4	0
1	6	4	7

Applications

□ Histogram processing is used in

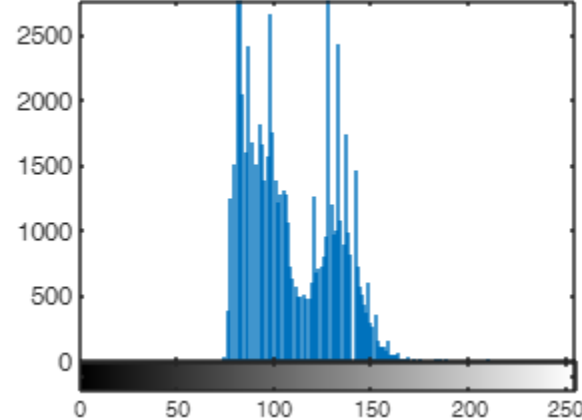
- Image Enhancement
- Image Segmentation
- Image Compression

Histogram Processing: Example 1

Original Image A



Histogram of Image A



Output Image sum



Output Image diff



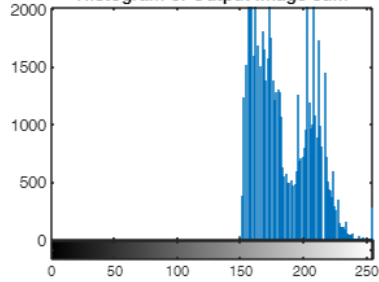
Output Image div



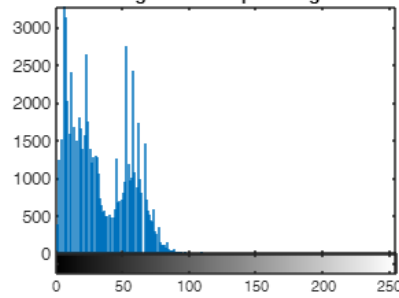
Output Image Heq



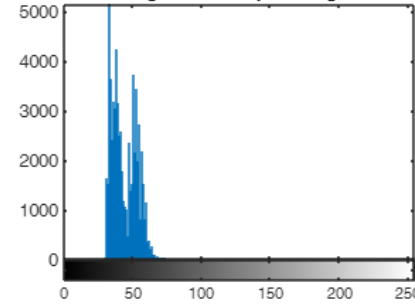
Histogram of Output Image sum



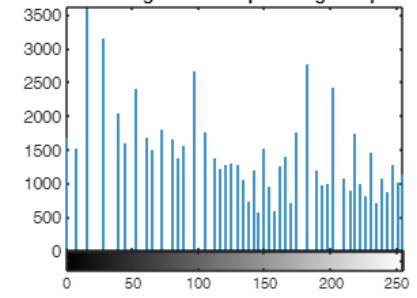
Histogram of Output Image diff



Histogram of Output Image div



Histogram of Output Image heq



Histogram Processing: Example 2

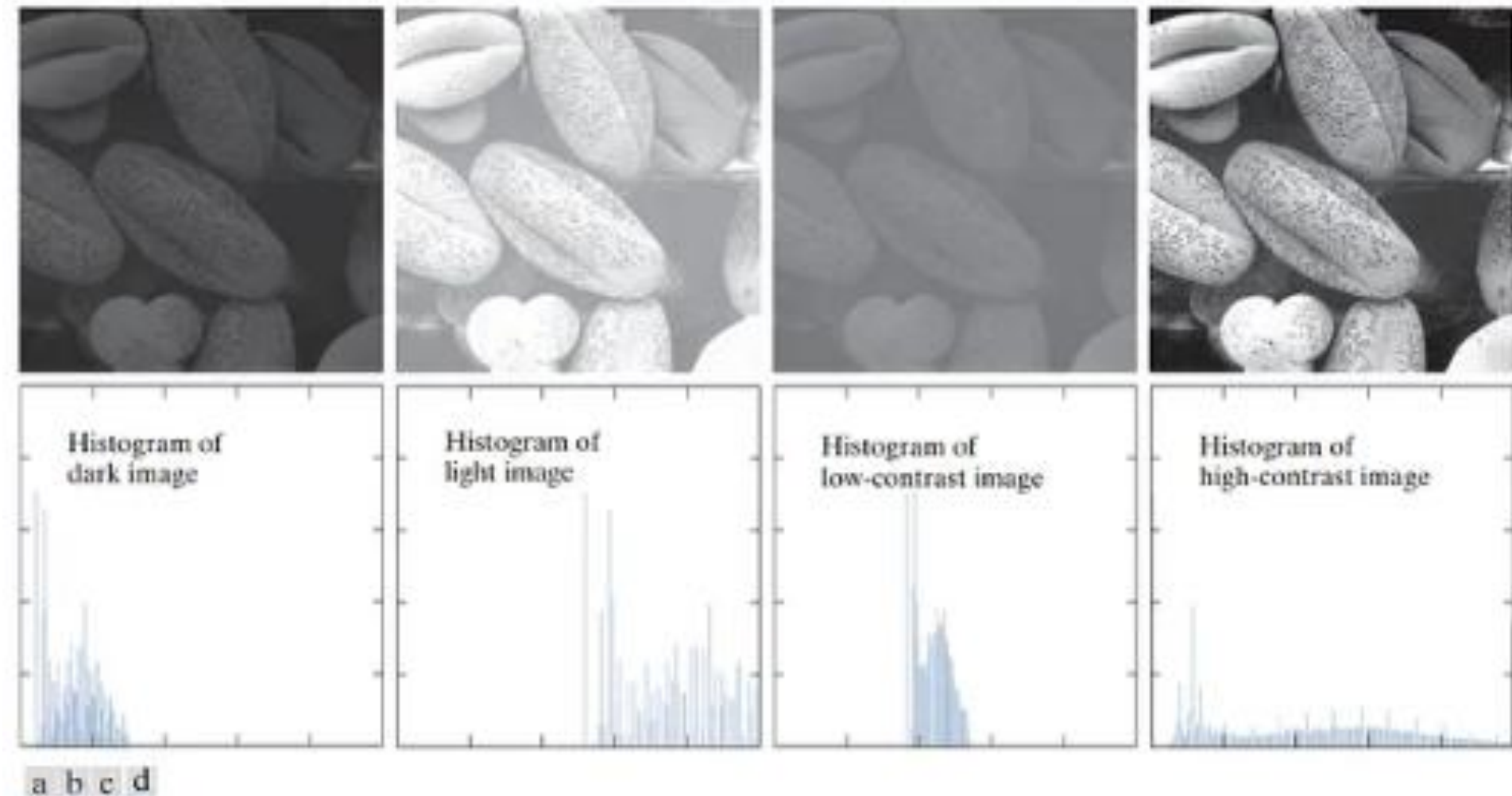
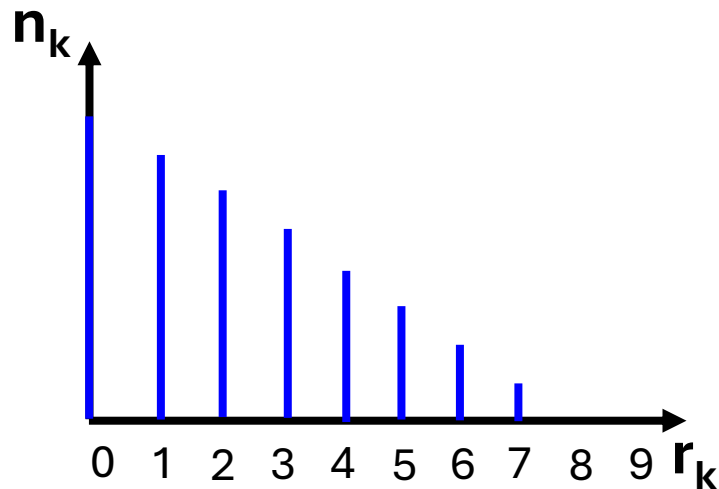


FIGURE 3.16 Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.

Histogram Equalization

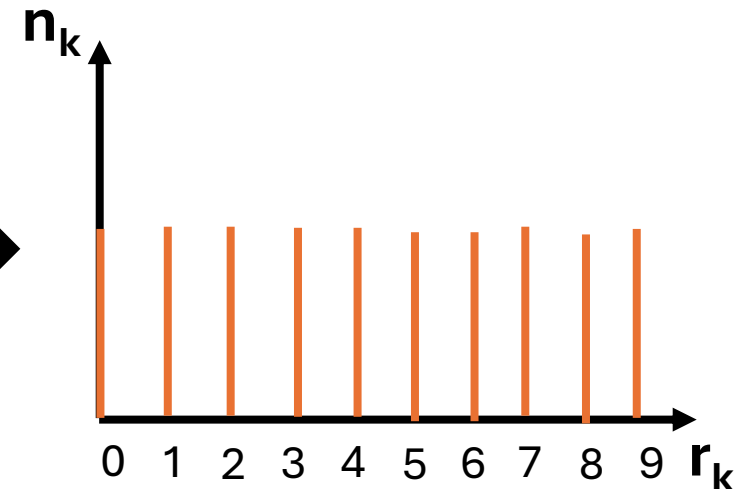
Histogram Equalisation

- It is the process that transforms the intensity values so that the histogram of the output image approximately matches the flat (uniform) histogram.



Darker Image
Poor Quality Image

Histogram
Equalisation



High Quality Image

Histogram Equalisation

- **Histogram Equalization:** it is a method to obtain a flatten histogram.
- The aim is to create an image with equally distributed brightness levels over the whole brightness scale.
- We can do it by adjusting the probability density function (pdf) of the original histogram of the image so that the probability spread equally
- Histogram Equalization automatically determines a transformation function to produce a new image with a uniform histogram

Histogram Equalisation

□ Condition for Histogram Equalisation:

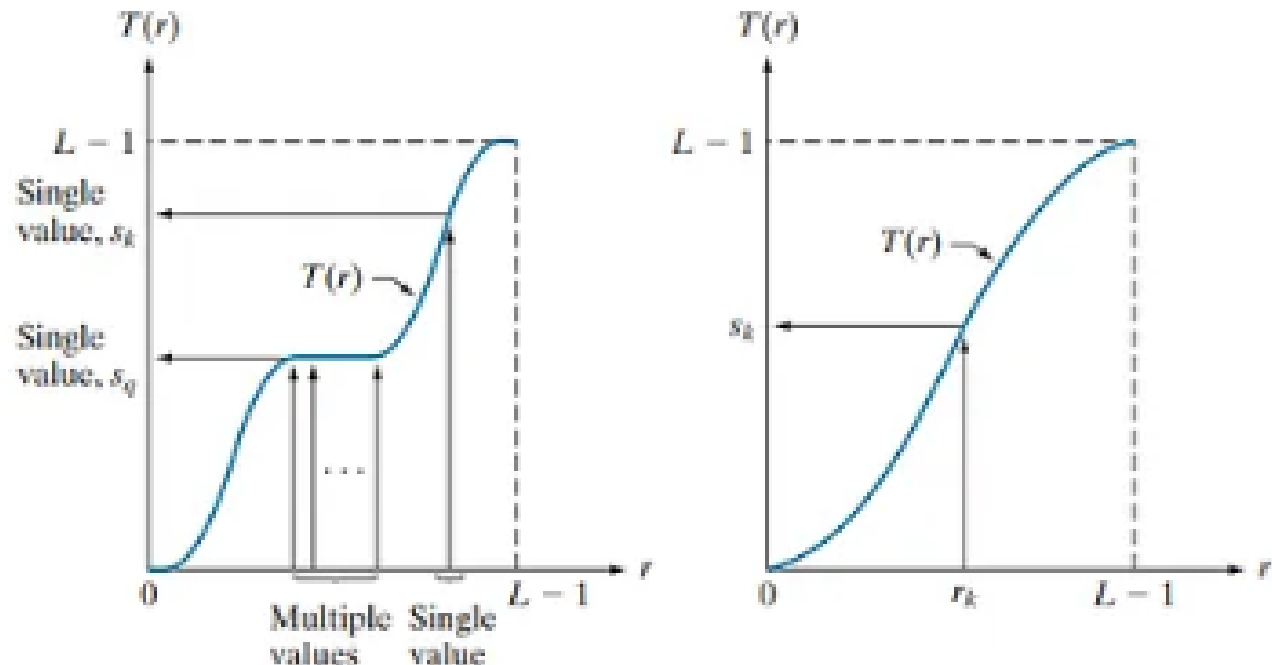
$$S = T(r)$$

- $T(r)$ is single-valued and monotonically increasing in interval $[0, 1]$
- $0 \leq T(r) \leq 1$ for $0 \leq r \leq 1$

a b

FIGURE 3.17

(a) Monotonic increasing function, showing how multiple values can map to a single value. (b) Strictly monotonic increasing function. This is a one-to-one mapping, both ways.



Numerical Example

TYPE 1

Ques 1: Suppose that a 3-bit image ($L = 8$) of size 64×64 pixels ($MN = 4096$) has the intensity distribution shown in following table. Get the histogram equalization transformation function and give the $p(s_k)$ for each s_k

r_k	0	1	2	3	4	5	6	7
n_k	790	1023	850	656	329	245	122	81

Numerical Example

i/p Gray Level (r_k)	no. of pixels (n_k)	$p(r_k) = n_k/MN$	Σ	$(L-1)\Sigma$	o/p Gray Level (s)
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7.00	7

Numerical Example

(r_k)	(n_k)	$p(r_k)$	Σ	$(L-1)\Sigma$	(s_k)
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7.00	7

Equalized	n_k
0	0
1	790
2	0
3	1023
4	0
5	850
6	$656 + 329 = 985$
7	$245 + 122 + 81 = 448$

Numerical Example

TYPE 2

Ques2: Apply the histogram Equalization technique to the following image

$f(x, y) =$

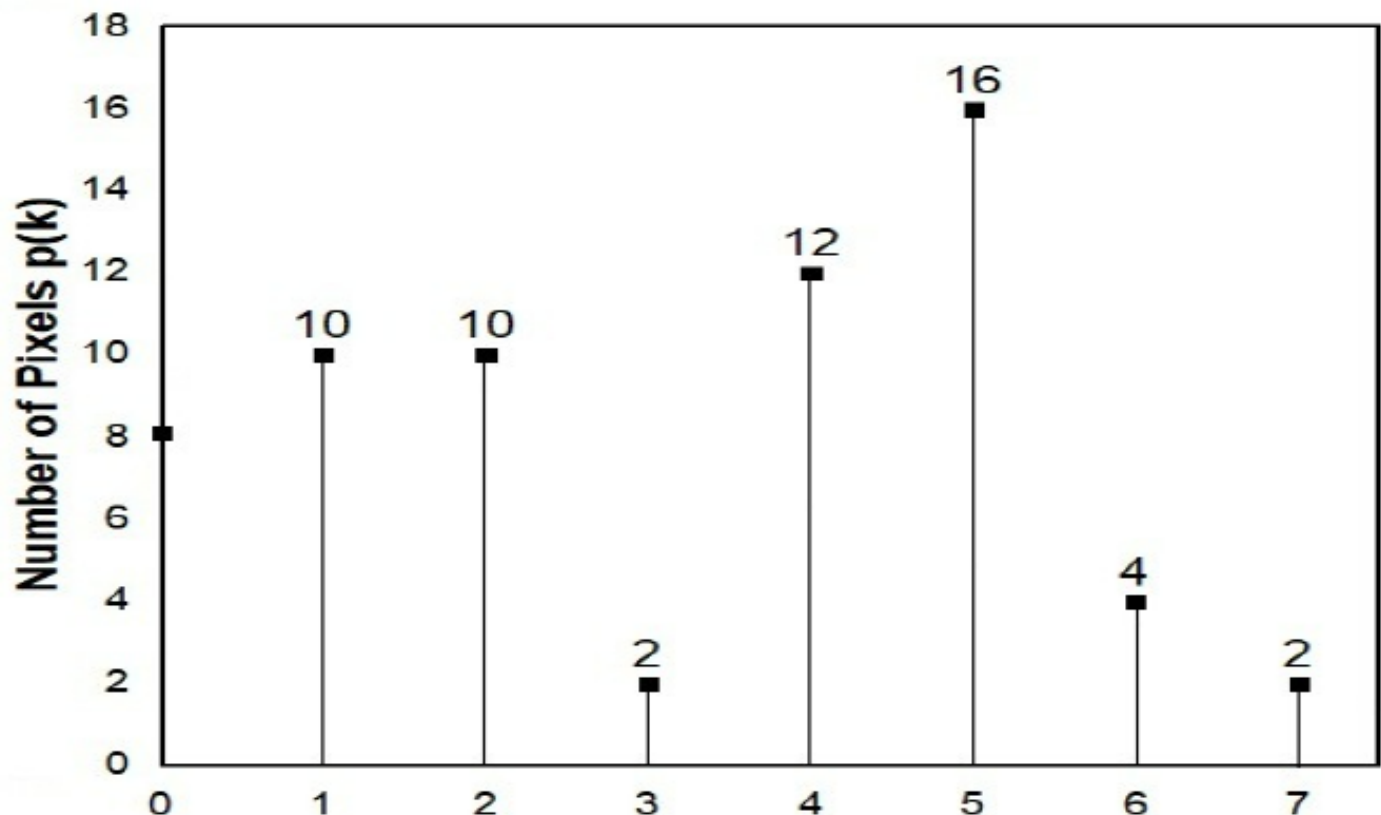
1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

Numerical Examples

Ques 8: Perform Histogram Equalisation for 8×8 eight level image described in Figure. Also, construct the histogram of the equalized image.

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TYPE 3



Histogram Equalisation

Disadvantages:

- Histogram equalization has a disadvantage that it can generate only one type of output image
- Sometimes it is useful to specify the shape of the histogram.
- To overcome this limitation, we prefer **Histogram Matching (Specification)**.
- With histogram specification, we can specify the shape of the histogram.
- It doesn't have to be a uniform histogram.

Histogram Matching (Specification)

Histogram Matching (Specification)

- Histogram equalization has a disadvantage that it can generate only one type of output image
- Sometimes it is useful to specify the shape of the histogram
- With histogram specification, we can specify the shape of the histogram
- It doesn't have to be a uniform histogram

□ For an image, whose enhancement is to be done, we are given an histogram, $G(z_k)$, that actually shows how the processed image's histogram should look after applying the transformation function to the i/p image

Numerical Examples

TYPE 1

Ques1: Perform histogram specification on the 8×8 , eight level image described in Table below

Table I: Pixel distribution of the image

i/p gray level (r_k)	0	1	2	3	4	5	6	7
No. of pixel (n_k)	8	10	10	2	12	16	4	2

Table II: Target Histogram

i/p gray level (r_k)	0	1	2	3	4	5	6	7
No. of pixel (n_k)	0	0	0	0	20	20	16	8

Numerical Examples

Solution:

Step 1: Perform Histogram Equalisation for Pixel distribution

r_k	n_k	$\text{pdf} = \frac{n_k}{N}$	$\text{cpdf} (\Sigma)$	$(L-1)\Sigma$	o/p gray level (s_k)
0	8	0.13	0.13	0.91	1
1	10	0.16	0.29	2.03	2
2	10	0.16	0.45	3.15	3
3	2	0.03	0.48	3.36	3
4	12	0.19	0.67	4.69	5
5	16	0.25	0.92	6.44	6
6	4	0.06	0.98	6.86	7
7	2	0.03	1.01	7.07	7

Total = 16

Numerical Examples

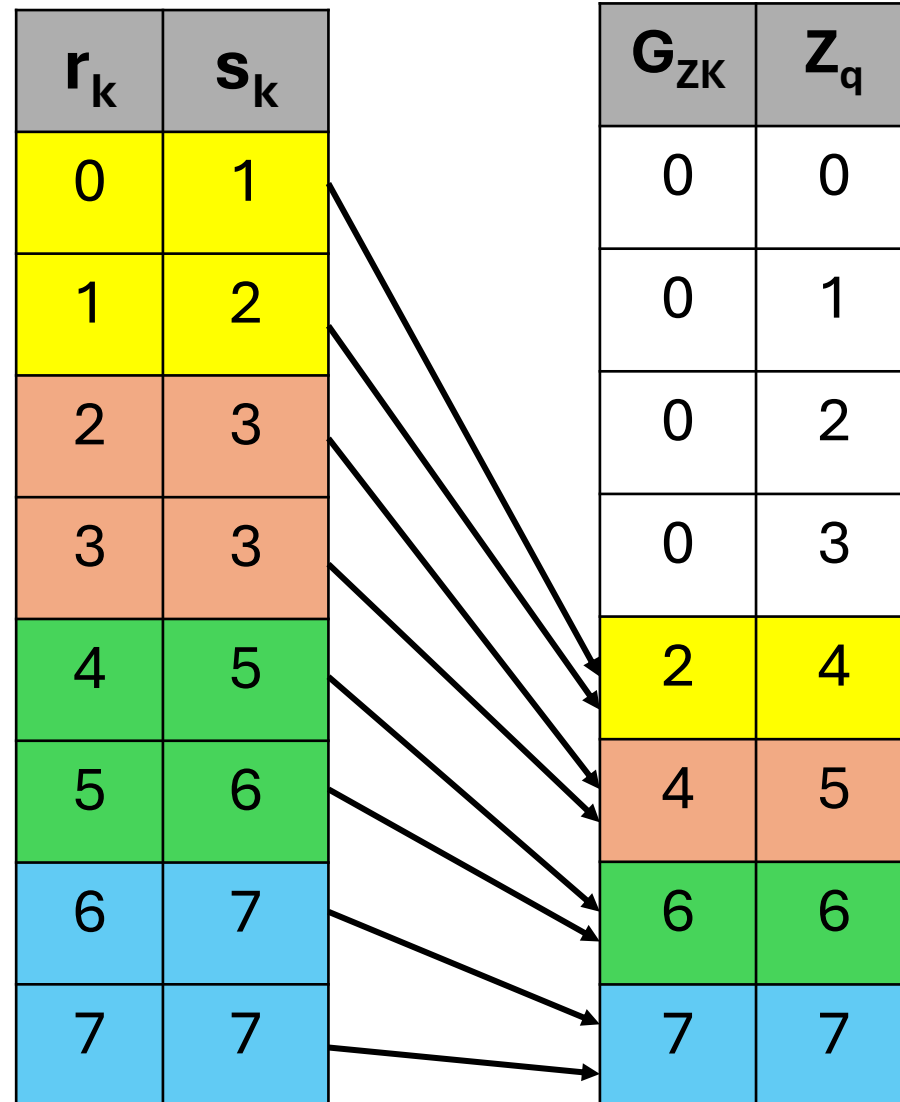
Step 2: Perform Histogram Equalisation for Target Histogram

Z_q	n_z	$\text{pdf} = \frac{n_z}{N}$	$\text{cpdf} (\Sigma)$	$(L-1)\Sigma$	o/p gray level (G_{zk})
0	0	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	20	0.31	0.31	2.17	2
5	20	0.31	0.62	4.34	4
6	16	0.25	0.87	6.09	6
7	8	0.13	1.00	7.00	7

Total = 64

Numerical Examples

Step 3: Mapping



Numerical Examples

Step 4: Final Mapping

r_k	s_k	G_{zk}	Map
0	1	0	4
1	2	0	4
2	3	0	5
3	3	0	5
4	5	2	6
5	6	4	6
6	7	6	7
7	7	7	7

Numerical Examples

Ques1: Suppose that a 3-bit image of size 64×64 pixels has the intensity distribution shown in the table (on the left). Get the histogram transformation function and make the output image with the specified histogram, listed in the table on the right

Table 1

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Table 2

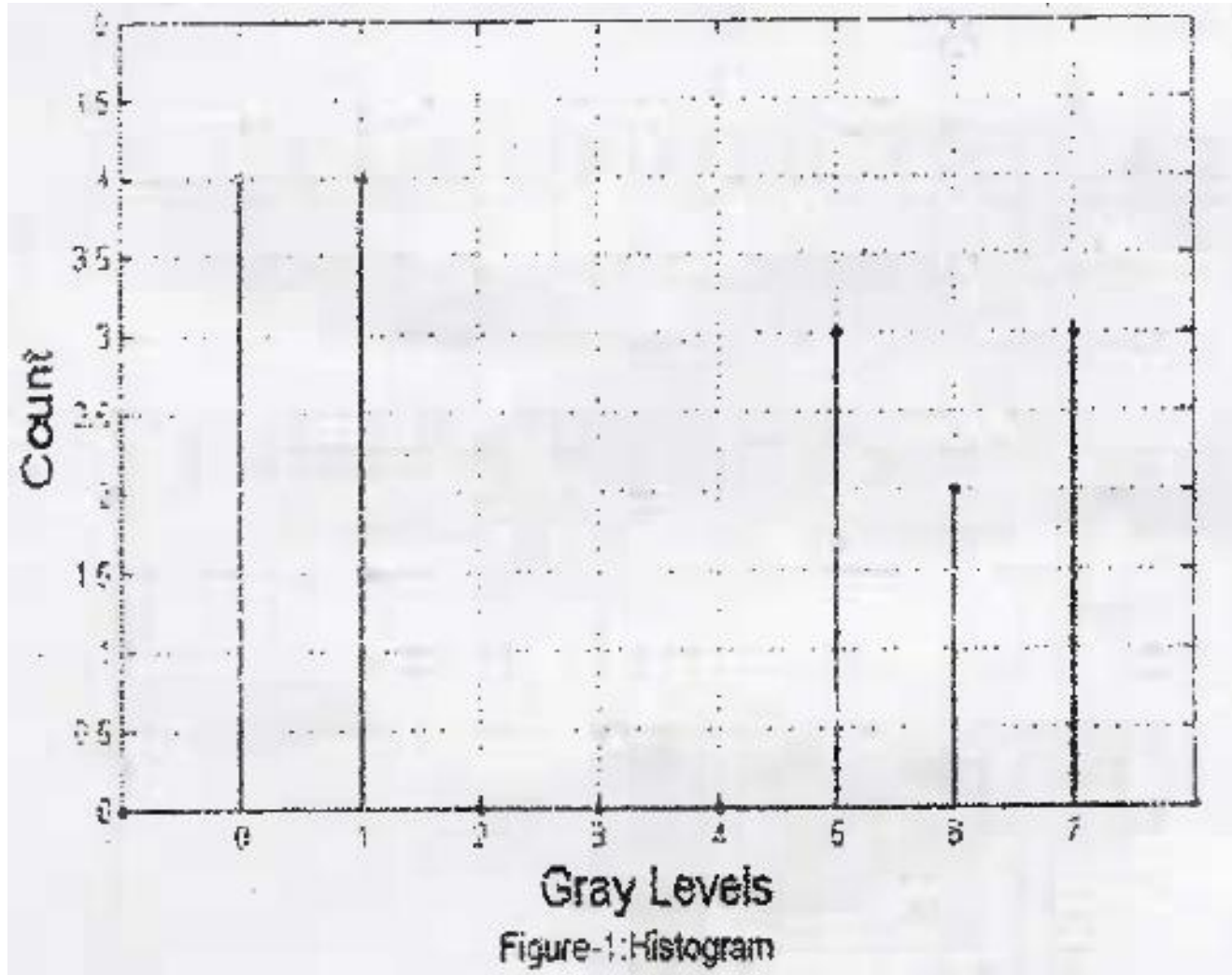
z_q	Specified $p_z(z_q)$
$z_0 = 0$	0.00
$z_1 = 1$	0.00
$z_2 = 2$	0.00
$z_3 = 3$	0.15
$z_4 = 4$	0.20
$z_5 = 5$	0.30
$z_6 = 6$	0.20
$z_7 = 7$	0.15

TYPE 2

Numerical Examples

Ques 3: Histogram of a 3-bits/pixel square image is given in figure 1,.

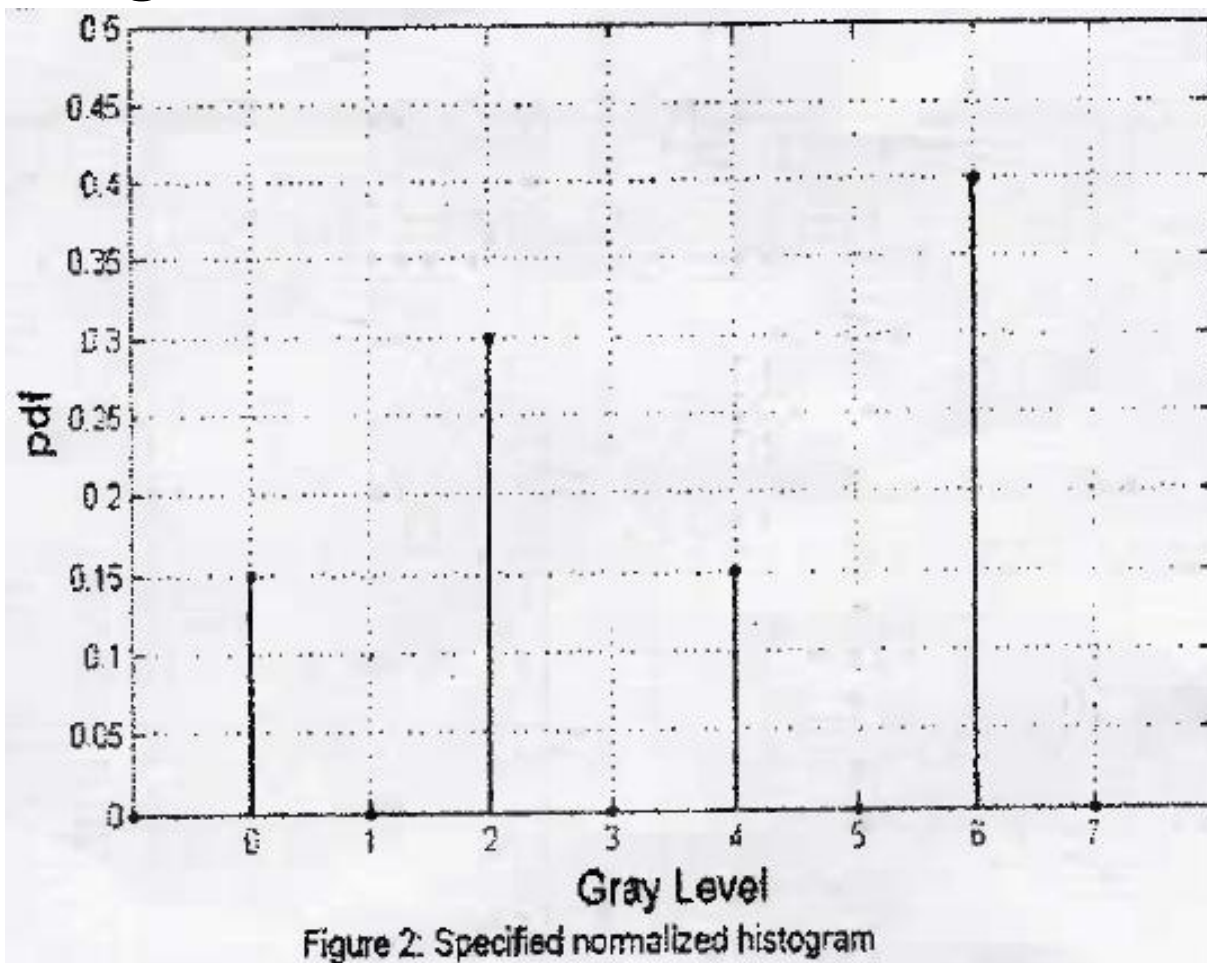
TYPE 3



Numerical Examples

specified normalised histogram (pdf) is given in figure 2.
Use histogram specification to generate histogram of
enhanced image

TYPE 3



Histogram Stretching

Histogram Stretching

- The aim of this transformation is to spread the histogram to cover the entire dynamic range, instead of changing the slope of the histogram.
- This operation is called as Histogram Stretching (Also known as Histogram Scaling).
- The mapping function for histogram stretching is given as follows:

$$\frac{S_{max} - S_{min}}{r_{max} - r_{min}}$$

- The transform is given as follows:

$$\left[\frac{S_{max} - S_{min}}{r_{max} - r_{min}} \right] (r - r_{min}) + S_{min}$$

Histogram Stretching

- The transform is given as follows:

$$\left[\frac{S_{max} - S_{min}}{r_{max} - r_{min}} \right] (r - r_{min}) + S_{min}$$

Where,

r_{max} , r_{min} : are the maximum and minimum gray-level values in the original image

S_{max} , S_{min} : are the maximum and minimum of the pixel values in the stretched histogram.

Numerical Examples

TYPE 1

Ques1: Perform histogram stretching on the 8×8 , eight level gray image, the gray level distribution of which is show in table. Perform the histogram stretching from gray level 0 to 7

r_k	0	1	2	3	4	5	6	7
n_k	0	0	5	20	20	19	0	0

Numerical Examples

TYPE 2

Ques2: A certain image has 10 Gray level intensities in the range 6 to 15. If we generate a linear contrast stretched image with min and max Gray level are 0 to 7 respectively. Write the formula and show the mapping in the tabular format as shown below.

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r	s	Round Off

Numerical Examples

TYPE 3

Ques3: Assume a 4-bit gray scale I as shown below. Generate the linear contrast stretched image with minimum gray level 5 and maximum gray level 14

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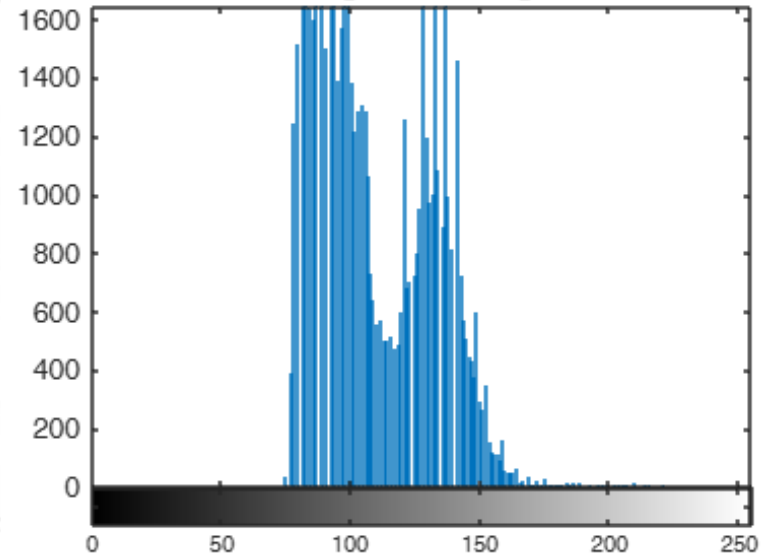
3	3	3	3	3	3	3	3
3	4	4	4	4	4	4	3
3	4	2	2	2	2	4	3
3	4	2	5	5	2	4	3
3	4	2	5	5	2	4	3
3	4	2	2	2	2	4	3
3	4	4	4	4	4	4	3
3	3	3	3	3	3	3	3

Histogram Stretching

Original Image A



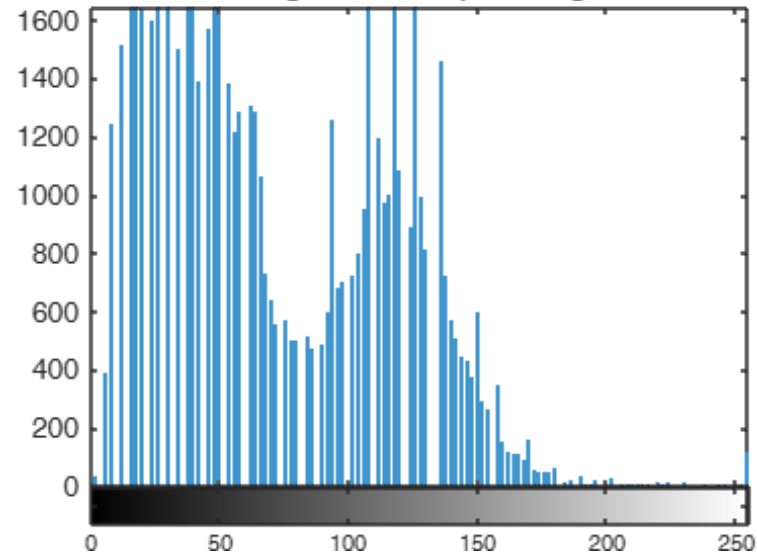
Histogram of Image A



Output Image B



Histogram of Output Image B



Histogram Statistics For Enhanced Image

Histogram Statistics

Let r :	be the discrete random variable representing the intensity values in the range $[0, L-1]$.
$P(r_i)$:	Normalised histogram (probability that the intensity r_i occurs in the image)

- The n^{th} moment (μ_n) of the mean 'm' is given as:

$$\mu_n(r) = \sum_{i=0}^{L-1} (r_i - m)^n P(r_i)$$

- Mean(m):

$$m = \sum_{i=0}^{L-1} r_i P(r_i)$$

- 2nd Moment (Also known as Variance):

$$\mu_2(r) = \sigma^2 = \sum_{i=0}^{L-1} (r_i - m)^2 P(r_i)$$

Histogram Statistics

❑ Alternative Approach:

- The sample mean and sample Variance can be expressed as:

$$m = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

$$\sigma^2 = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x, y) - m]^2$$

Where,

$x = 0, 1, 2, \dots, M-1$ (represents number of rows)

$y = 0, 1, 2, \dots, N-1$ (represent number of columns)

Numerical Examples

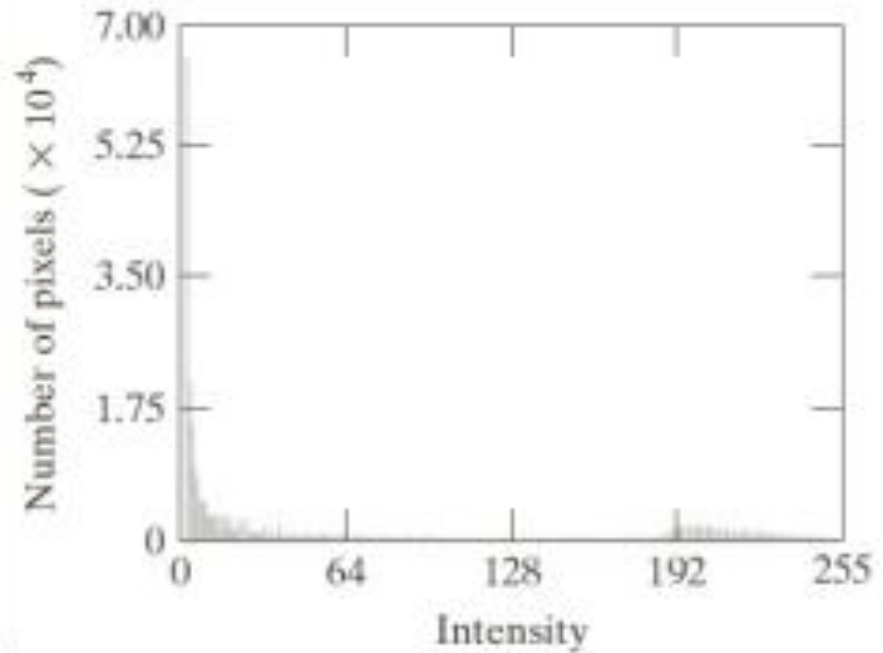
Ques1: Consider the following 2-bit image of size 5×5

0	0	1	1	2
1	2	3	0	1
3	3	2	2	0
2	3	1	0	0
1	1	3	2	2

Determine the (A) Mean and (B) Variance.

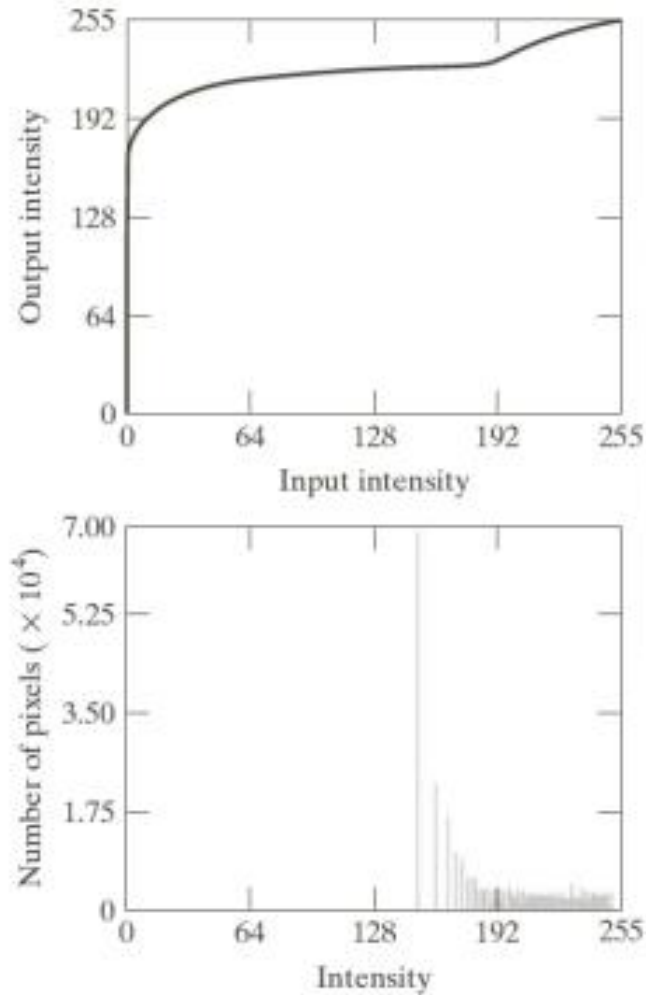
Histogram Equalization Vs Histogram Specification

Original Image



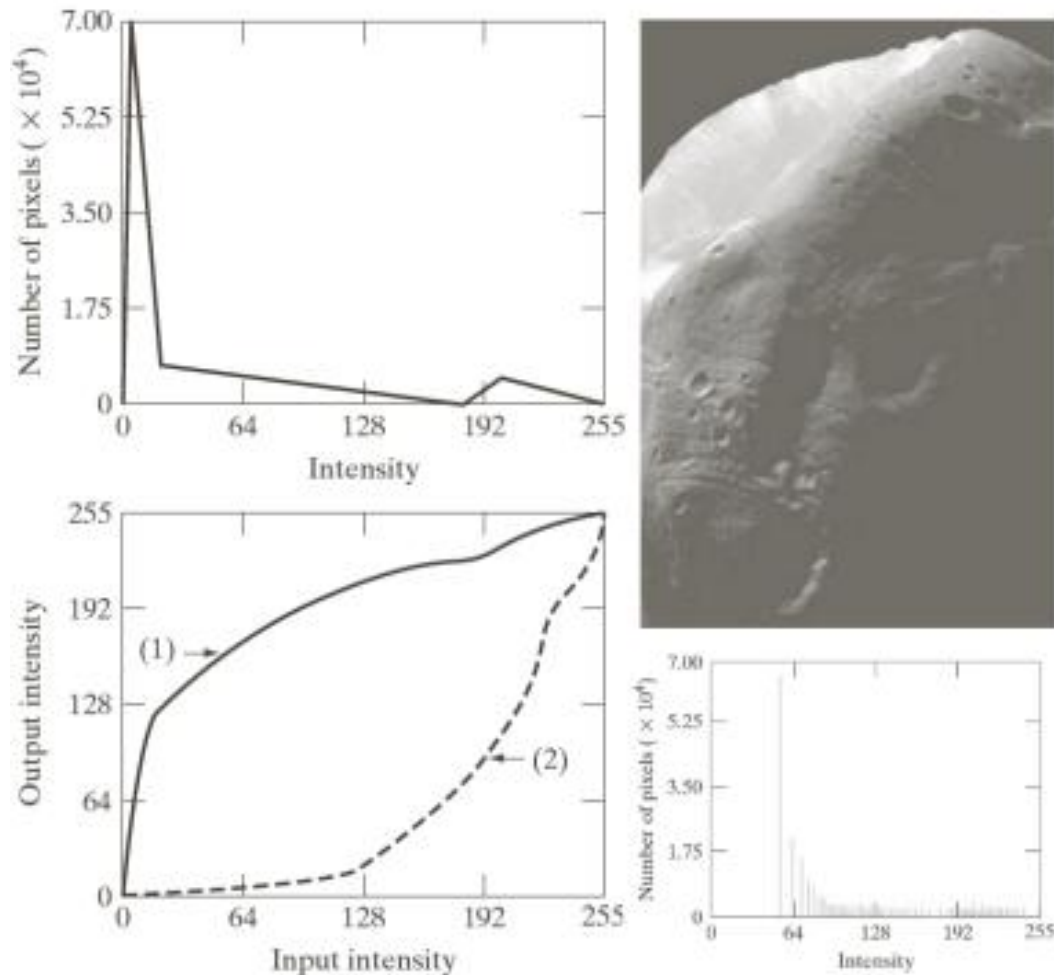
Large concentration of pixels in the dark region of the histogram.

After Histogram Equalisation



Processed image after Histogram Equalisation, is an enhanced image over original image. However, the histogram of processed image lies in the Gray level range of 128 and above.

After Histogram Specification

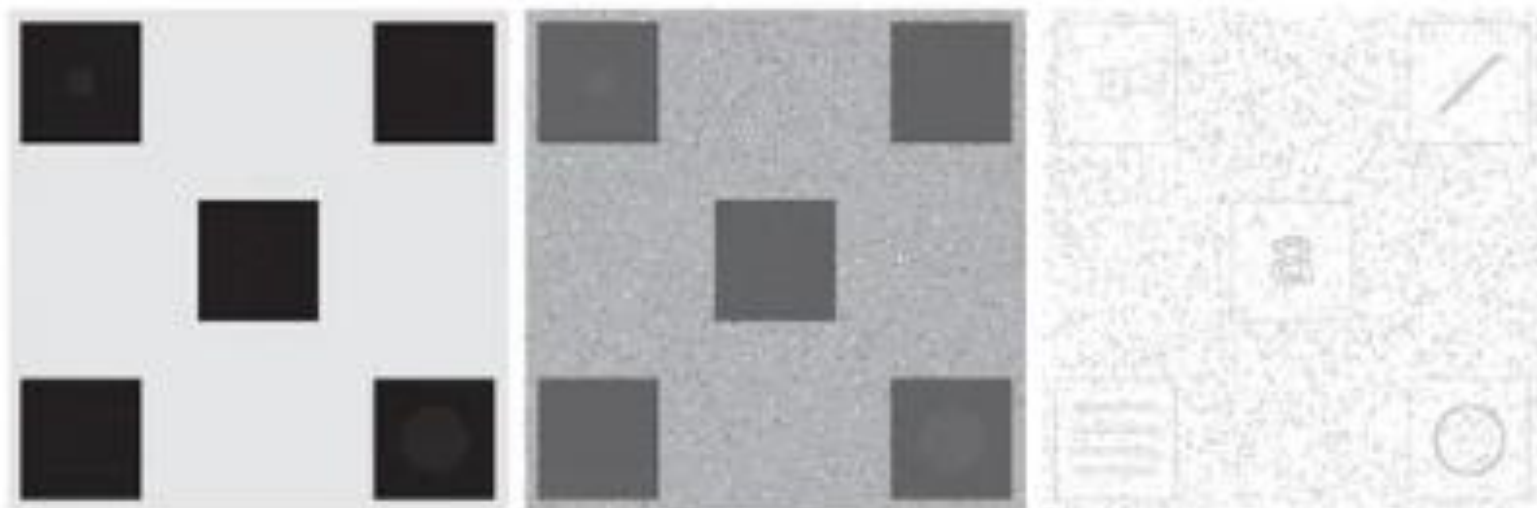


Processed image after Histogram Specification, is an enhanced image over original image and Histogram Equalisation. Moreover, the histogram of processed image cover the entire Gray scale.

Local Histogram Processing

Local Histogram Processing

a b c
FIGURE 3.26
(a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization.



Local Histogram Processing

- The histogram processing methods discussed thus far are global, in the sense that pixels are modified by a transformation function based on the intensity distribution of an entire image.
- This global approach is suitable for overall enhancement, but generally fails when the objective is to enhance details over small areas in an image,
- This is because the number of pixels in small areas have negligible influence on the computation of global transformation.
- The solution is to devise transformation functions based on the intensity distribution of pixel neighbourhood. The histogram processing techniques previously described can be adapted to local enhancement.

Local Histogram Processing

- The procedure is to define a neighbourhood and move its center from pixel to pixel in a horizontal or vertical direction.
- At each location, the histogram of the points in the neighbourhood is computed, and either a histogram equalisation or histogram specification transformation function is obtained.
- This function is used to map the intensity of the pixel centered in the neighbourhood. The center of the neighbourhood is then moved to an adjacent pixel location and the procedure is repeated.



Thank You